Ch 3.1: Linear Regression

Lecture 4 - CMSE 381

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Fri, Sep 9, 2022

Announcements

Last time:

- 2.2 Assessing Model Accuracy

Announcements:

- Office Hours
- Homework - Due Wed 9/14
  - ↳ Crowdmark
  - ↳ Get an email
  - ↳ Sep. file for each question

Covered in this lecture

- Least squares coefficient estimates for linear regression
- Residual sum of squares (RSS)
- Confidence interval, hypothesis test, and p-value for coefficient estimates
- Residual standard error (RSE)
- R squared

Section 1

Simple Linear Regression

Setup

Regression
Y is quantitative variable

- Predict Y on a single predictor variable X

quantitative variable
$p = 1$

$$Y \approx \underbrace{\beta_0 + \beta_1}_{\text{coefficients}} X$$

- "$\approx$" .... "is approximately modeled as"

β_1 = slope
β_0 = intercept

$y = mx + b$

Example

X Y

		TV	Radio	Newspaper	Sales
1					
2	1	230.1	37.8	69.2	22.1
3	2	44.5	39.3	45.1	10.4
4	3	17.2	45.9	69.3	9.3
5	4	151.5	41.3	58.5	18.5
6	5	180.8	10.8	58.4	12.9
7	6	8.7	48.9	75	7.2
8	7	57.5	32.8	23.5	11.8
9	8	120.2	19.6	11.6	13.2
10	9	8.6	2.1	1	4.8
11	10	199.8	2.6	21.2	10.6
12	11	66.1	5.8	24.2	8.6

$$Y \approx \beta_0 + \beta_1 X$$

$$\text{Sales} \approx \beta_0 + \beta_1 \cdot \text{TV}$$

intercept slope

Parameters / coefficients to be learned

Once I've chosen guess for β_0, β_1 model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

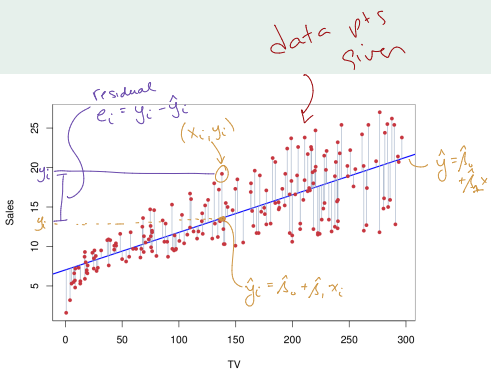
Least squares criterion: Setup

How do we estimate the coefficients?

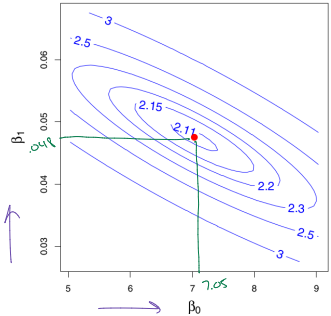
Given $\{(x_1, y_1), \ldots, (x_n, y_n)\}$

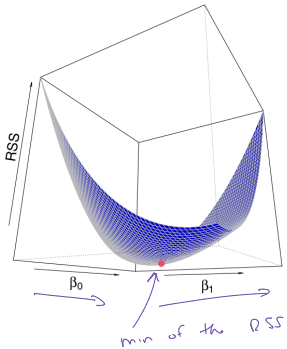
Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

$e_i = y - y_i$ is the ith residual



Least squares criterion: RSS





Residual sum of squares RSS is

$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$
$$= \sum_i (y_i - \hat{y}_i)^2$$

$$\text{sales} \approx \beta_0 + \beta_1 \text{TV}$$

later, this will be our coefficients

Least squares criterion

Find β_0 and β_1 that minimize the RSS.

Least squares coefficient estimates

$$\min_{\hat{\beta}_0,\hat{\beta}_1} \sum_i (y_i - \hat{y}_i)^2$$

$$(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

$$\frac{\partial RSS}{\partial \beta_0} = -2\sum_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\frac{\partial RSS}{\partial \beta_1} = -2\sum_i x_i(y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

avg of y_i

avg of x_i

$$\overline{y} = \frac{1}{n}\sum_{i=1}^n y_i$$

Coding group work

Section 2

Assessing Coefficient Estimate Accuracy

Bias in estimation

Analogy with mean

- Assume a true value μ^*
- An estimate from training data $\hat{\mu}$
- The estimate is unbiased if $E(\hat{\mu} = \mu^*)$

- Sample mean is unbiased for population mean:

$$E(\hat{\mu}) = E\left(\frac{1}{n}\sum_i X_i\right) = \mu^*$$

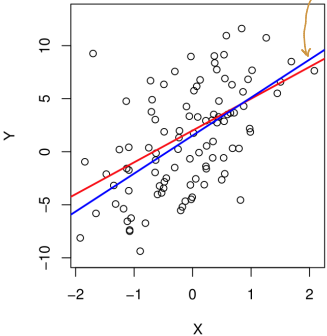
- Standard variance estimate is biased

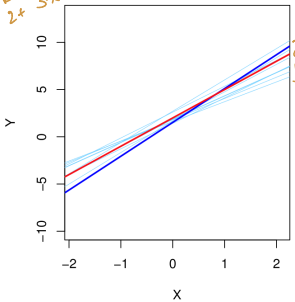
$$E(\hat{\sigma}^2) = E\left[\frac{1}{n}\sum_i (X_i - \overline{X})^2\right] \neq \sigma^2$$

Linear regression is unbiased

least squares estimate

Red true line
2+ 3x





Blue least squares estimates for diff. samples of data

100 data pts from

$$y = 2 + 3x + \varepsilon$$

normal dist w/ mean 0

Coding group work

Run the section titled "Simulating data"

Variance in estimation

Continuing analogy with mean

- True value μ^*
- Estimate from training data $\hat{\mu}$
- Variance of sample mean
  $\mathrm{Var}(\hat{\mu}) = \underbrace{\mathrm{SE}(\hat{\mu})}_{\text{Standard error}}{}^2 = \frac{\sigma^2}{n}$

More data
$n \to \infty$

Means standard error is smaller

Means better estimate

Variance of linear regression estimates

- Variance of linear regression estimates:

???

$$\mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

Replace $\sigma^2 \rightsquigarrow \hat{\sigma}^2$

estimate $\widehat{\mathrm{SE}}(\hat{\beta}_0)$

We're lazy and drop the top hat

- Residual standard error is an estimate of σ

$$RSE = \sqrt{RSS/(n-2)}$$

$\hat{\sigma} =$

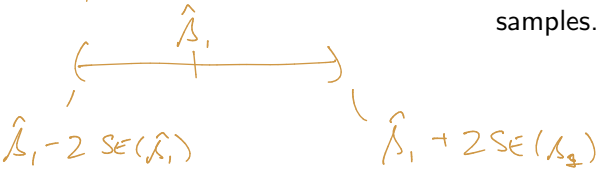
Residual sum of squares $\sum e_i^2 = \sum (y_i - \hat{y}_i)^2$

Confidence Interval

The 95% confidence interval for β_1 approximately takes the form

Some small terms we'll ignore

$$\hat{\beta}_1 \pm 2 \cdot \underbrace{\mathrm{SE}(\hat{\beta}_1)}$$



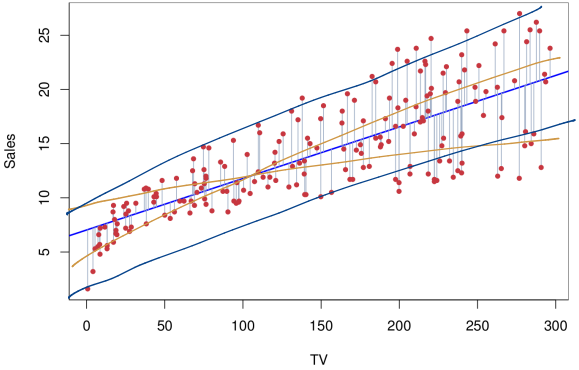
Interpretation:
There is approximately a 95% chance that the interval

$$\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$$

will contain β_1 where we repeatedly approximate $\hat{\beta}_1$ using repeated samples.

CI in Advertising data

Confidence Interval



For the advertising data set, the 95% CIs are:

- β_1 :: $[0.042, 0.053]$
- β_0 :: $[6.130, 7.935]$

Hypothesis testing

H_0: There is no relationship between X and Y

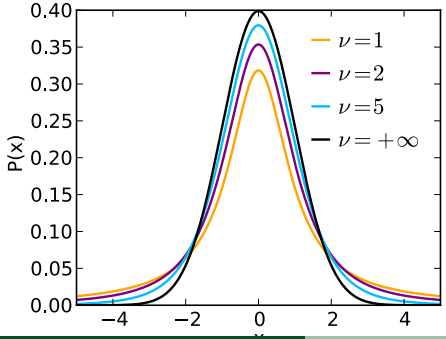
H_1: There is some relationship between X and Y

Test statistic and p-value

Test statistic:

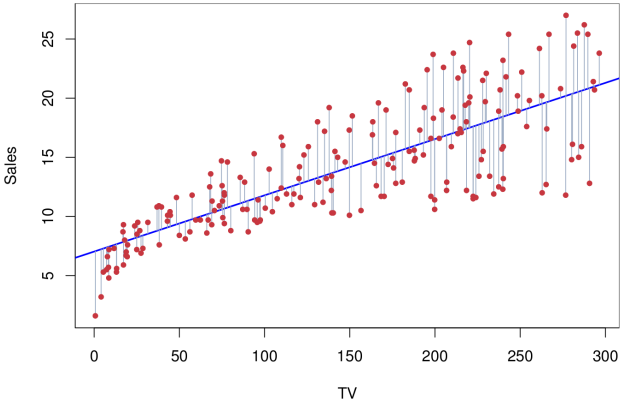
$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$$

t-distribution with $n - 2$ degrees of freedom



Advertising example

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	7.0325	0.4578	15.36	< 0.0001
`TV`	0.0475	0.0027	17.67	< 0.0001



Assessing the accuracy of the module: RSE

Residual standard error (RSE):

$$
\begin{aligned}
RSE &= \sqrt{\frac{1}{n-2} RSS} \\
&= \sqrt{\frac{1}{n-2} \sum_i (y_i - \hat{y}_i)^2}
\end{aligned}
$$

Assessing the accuracy of the module: R^2

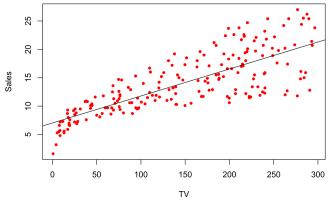
R squared:

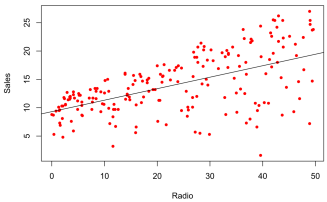
$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

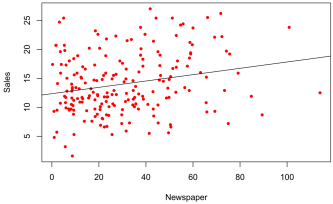
where total sum of squares is

$$TSS = \sum_i (y_i - \overline{y})^2$$

Advertising example







$R^2 = 0.61$ $R^2 = 0.33$ $R^2 = 0.05$

Coding group work

Run the section titled "Assessing Coefficient Estimate Accuracy"

Next time

Lec #	Date	Topic	Reading	Homeworks
1	W Aug 31	Intro / First day stuff / Python Review Pt 1	1	
2	F Sep 2	What is statistical learning? / Python Review Pt 2	2.1	
	M Sep 5	No class - Labor day		
3	W Sep 7	Assessing Model Accuracy	2.2	HW #1 Due
4	F Sep 9	Linear Regression	3.1	
5	M Sep 12	More Linear Regression	3.1, 3.2	
6	W Sep 14	Even more linear regression	3.3	HW #2 Due
7	F Sep 16	Probably more linear regression		
8	M Sep 19	Intro to classification, Logisitic Regression	4.1, 4.2, 4.3	
9	W Sep 21	More logistic regression		HW #3 Due
10	F Sep 23	Review		
11	M Sep 26	***Midterm #1***		
12	W Sep 28	[No class, Dr Munch out of town]		
13	F Sep 30	[No class, Dr Munch out of town]		
14	M Oct 3	Leave one out CV	5.1.1, 5.1.2	
15	W Oct 5	k-fold CV	5.1.3	
16	F Oct 7	More k-fold CV	5.1.4	
17	M Oct 10	CV for classification	5.1.5	HW #4 Due
18	W Oct 12	Resampling methods: Bootstrap	5.2	
19	F Oct 14	Subset selection	6.1	
20	M Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
21	W Oct 19	Shrinkage: Lasso	6.2.2	
22	F Oct 21	Dimension Reduction	6.3	

Announcements

- We had a quiz last time!
- Homework 2
  - NEW: Upload to crowdmark
  - Due Weds, Sep 14
  - Need to upload individual file for EACH QUESTION